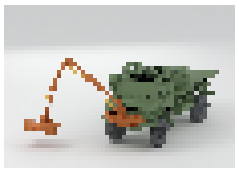
LEARNED-MODEL DATASETS

inverse dynamics · ID | forward dynamics · FD | reduced order · ROM

CLOSED-LOOP CAMPAIGN · R1 / R3



realistic operation

controller + waypoints

gravity-loaded motion

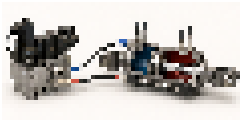
01 12-D Latin-hypercube randomization

02 1,000-run four-stage curriculum

03 state · pressure · command · descriptor

ALIGN + JOIN

OPEN-LOOP CAMPAIGN · R2



causal excitation

64 s command schedule

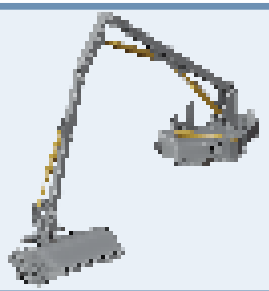
four randomized draws

01 single-channel + simultaneous phases

02 stroke-coverage acceptance gate

rejected export · duplicate-state audit

AUDIT · VALIDATION



SHARED SIMSCAPE DIGITAL PLANT

electro-hydraulic subsystem multibody mechanism

one plant interface · one time base · one schema

CAD / URDF geometry

ode23t stiff integration

nominal parameter set